

Q1.

The discrete random variable Y has the probability function

$$P(Y = y) = \begin{cases} 2ky & y = 1, 2, 3, 4 \\ 0 & \text{otherwise} \end{cases}$$

where k is a constant.

Show that $\text{Var}(5Y - 2) = 25$

(Total 6 marks)

Q2.

The random variable T can take the value $T = -2$ or any value in the range $0 \leq T < 12$

The distribution of T is given by $P(T = -2) = c$, $P(0 \leq T \leq t) = 225k - k(15 - t)^2$

(a) (i) Show that $1 - c = 216k$

(3)

(ii) Given that $c = 0.1$, find the value of $E(T)$

(3)

(b) Show that $E(\sqrt{|T|}) = \frac{5\sqrt{2} + 52\sqrt{3}}{50}$

(3)

(Total 9 marks)

Q3.

Aiden takes his car to a garage for its MOT test. The probability that his car will need to have X tyres replaced is shown in the table.

x	0	1	2	3	4
$P(X = x)$	0.1	0.35	0.25	0.2	0.1

(a) Show that the mean of X is 1.85 and calculate the variance of X .

(4)

(b) The charge for the MOT test is $\pounds c$ and the cost of **each** new tyre is $\pounds n$. The total amount that Aiden must pay the garage is $\pounds T$.

(i) Express T in terms of c , n and X .

(1)

(ii) Hence, using your results from part (a), find expressions for $E(T)$ and $\text{Var}(T)$.

(4)

(Total 9 marks)

Q4.

In a computer game, players try to collect five treasures. The number of treasures that Isaac collects in one play of the game is represented by the discrete random variable X .

The probability distribution of X is defined by

$$P(X = x) = \begin{cases} \frac{1}{x+2} & x = 1, 2, 3, 4 \\ k & x = 5 \\ 0 & \text{otherwise} \end{cases}$$

- (a) (i) Show that $k = \frac{1}{20}$. (2)
- (ii) Calculate the value of $E(X)$. (2)
- (iii) Show that $\text{Var}(X) = 1.5275$. (3)
- (iv) Find the probability that Isaac collects more than 2 treasures. (2)
- (b) The number of points that Isaac scores for collecting treasures is Y where
- $$Y = 100X - 50$$
- Calculate the mean and the standard deviation of Y . (4)

(Total 13 marks)

Q5.

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- (a) A discrete random variable X has a probability function defined by

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!} \text{ for } x = 0, 1, 2, 3, 4, \dots$$

- (i) State the name of the distribution of X . (1)
- (ii) Write down, in terms of λ , expressions for $E(3X - 1)$ and $\text{Var}(3X - 1)$. (2)
- (iii) Write down an expression for $P(X = x + 1)$, and hence show that
- $$P(X = x + 1) = \frac{\lambda}{x + 1} P(X = x)$$
- (3)
- (b) The number of cars and the number of coaches passing a certain road junction may

be modelled by independent Poisson distributions.

- (i) On a winter morning, an average of 500 cars per hour and an average of 10 coaches per hour pass this junction.

Determine the probability that a total of at least 10 such vehicles pass this junction during a particular 1-minute interval on a winter morning.

(3)

- (ii) On a summer morning, an average of 836 cars per hour and an average of 22 coaches per hour pass this junction.

Calculate the probability that a total of at most 3 such vehicles pass this junction during a particular 1-minute interval on a summer morning. Give your answer to two significant figures.

(3)

(Total 12 marks)

Q6.

A house has a total of five bedrooms, at least one of which is always rented.

The probability distribution for R , the number of bedrooms that are rented at any given time, is given by

$$P(R = r) = \begin{cases} 0.5 & r = 1 \\ 0.4(0.6)^{r-1} & r = 2, 3, 4 \\ 0.0296 & r = 5 \end{cases}$$

- (a) Complete the table below.

r	1	2	3	4	5
$P(R = r)$	0.5				0.0296

(2)

- (b) Find the probability that fewer than 3 bedrooms are **not** rented at any given time.

(3)

- (c) (i) Find the value of $E(R)$.

(2)

- (ii) Show that $E(R^2) = 4.8784$ and hence find the value of $\text{Var}(R)$.

(3)

- (d) Bedrooms are rented on a monthly basis.

The monthly income, $\pounds M$, from renting bedrooms in the house may be modelled by

$$M = 1250R - 282$$

Find the mean and the standard deviation of M .

(3)

(Total 13 marks)

Q7.

A small supermarket has a total of four checkouts, at least one of which is always staffed. The probability distribution for R , the number of checkouts that are staffed at any given time, is

$$P(R = r) = \begin{cases} \frac{2}{3} \left(\frac{1}{3}\right)^{r-1} & r = 1, 2, 3 \\ k & r = 4 \end{cases}$$

(a) Show that $k = \frac{1}{27}$.

(2)

(b) Find the probability that, at any given time, there will be at least 3 checkouts that are staffed.

(1)

(c) It is suggested that the total number of customers, C , that can be served at the checkouts per hour may be modelled by

$$C = 27R + 5$$

Find:

(i) $E(C)$;

(3)

(ii) the standard deviation of C .

(4)

(Total 10 marks)

Q8.

Joanne has 10 identically-shaped discs, of which 1 is blue, 2 are green, 3 are yellow and 4 are red. She places the 10 discs in a bag and asks her friend David to play a game by selecting, at random and without replacement, two discs from the bag.

(a) Show that:

(i) the probability that the two discs selected are the same colour is $\frac{2}{9}$;

(2)

(ii) the probability that exactly one of the two discs selected is blue is $\frac{1}{5}$.

(2)

- (b) Using the discs, Joanne plays the game with David, under the following conditions:

If the two discs selected by David are the same colour, she will pay him 135p. If exactly one of the two discs selected by David is blue, she will pay him 145p. Otherwise David will pay Joanne 45p.

- (i) When a game is played, X is the amount, in pence, **won by David**. Construct the probability distribution for X , in the form of a table.

(2)

- (ii) Show that $E(X) = 33$.

(2)

- (c) Joanne modifies the game so that the amount per game, Y pence, that **she wins** may be modelled by

$$Y = 104 - 3X$$

- (i) Determine how much Joanne would expect to win if the game is played 100 times.

(3)

- (ii) Calculate the standard deviation of Y , giving your answer to the nearest 1p.

(4)

(Total 15 marks)

Q9.

- (a) The number of text messages, N , sent by Peter each month on his mobile phone never exceeds 40.

When $0 \leq N \leq 10$, he is charged for 5 messages.

When $10 < N \leq 20$, he is charged for 15 messages.

When $20 < N \leq 30$, he is charged for 25 messages.

When $30 < N \leq 40$, he is charged for 35 messages.

The number of text messages, Y , that Peter is charged for each month has the following probability distribution:

y	5	15	25	35
$P(Y = y)$	0.1	0.2	0.3	0.4

- (i) Calculate the mean and the standard deviation of Y .

(4)

- (ii) The Goodtime phone company makes a total charge for text messages, C pence, each month given by

$$C = 10Y + 5$$

Calculate $E(C)$.

(1)

- (b) The number of text messages, X , sent by Joanne each month on her mobile phone is such that

$$E(X) = 8.35 \quad \text{and} \quad E(X^2) = 75.25$$

The Newtime phone company makes a total charge for text messages, T pence, each month given by

$$T = 0.4X + 250$$

Calculate $\text{Var}(T)$.

(4)

(Total 9 marks)

Q10.

The number of mistakes, X , that Holli makes as a learner driver when she drives from Ampthill to Bedford can be modelled by the following discrete probability distribution:

x	≤ 1	2	3	4	5	6	≥ 7
$P(X = x)$	0	0.40	0.25	0.18	0.12	k	0

- (a) Find the value of k .

(1)

- (b) Find:

(i) $E(X)$;

(1)

(ii) $\text{Var}(X)$

(3)

- (c) When Holli makes the return journey by the same route, the number of mistakes, Y , that she makes can be approximated by

$$Y = 2X - 3$$

Find:

(i) $E(Y)$;

(1)

- (ii) the standard deviation of Y .

(3)

(Total 9 marks)

Q11.

The number of fish, X , caught by Pearl when she goes fishing can be modelled by the following discrete probability distribution:

x	1	2	3	4	5	6	≥ 7
$P(X = x)$	0.01	0.05	0.14	0.30	k	0.12	0

- (a) Find the value of k . (1)
- (b) Find:
- (i) $E(X)$; (1)
- (ii) $\text{Var}(X)$. (3)
- (c) When Pearl sells her fish, she earns a profit, in pounds, given by
- $$Y = 5X + 2$$
- Find:
- (i) $E(Y)$; (1)
- (ii) the standard deviation of Y . (3)
- (Total 9 marks)

Q12.

The Globe Express agency organises trips to the theatre. The cost, $\pounds X$, of these trips can be modelled by the following probability distribution:

x	40	45	55	74
$P(X = x)$	0.30	0.24	0.36	0.10

- (a) Calculate the mean and standard deviation of X . (4)
- (b) For special celebrity charity performances, Globe Express increases the cost of the trips to $\pounds Y$, where

$$Y = 10X + 250$$

Determine the mean and standard deviation of Y .

(2)
(Total 6 marks)

Q13.

Boxes contain screws of varying nominal lengths.

The number of screws with nominal lengths less than 2 inches in a box may be modelled by a discrete random variable, L , with mean 580 and variance 1600.

The number of screws with nominal lengths greater than 2 inches in a box may be modelled by a discrete random variable, G , defined by:

$$G = \frac{L}{2} + 80.$$

- (a) Find values for the mean and variance of G .

(3)

- (b) Each box contains 1000 screws.

- (i) Deduce, in terms of L , an expression for the number, R , of screws with a nominal length of exactly 2 inches in a box.

(2)

- (ii) Hence determine values for the mean and variance of R .

(3)

(Total 8 marks)

Q14.

The probability distribution for the number, R , of unwrapped sweets in a tin is given in the following table.

r	0	1	2	3	4
$P(R = r)$	0.1	0.2	0.4	0.2	0.1

- (a) Show that:

- (i) $E(R) = 2$;

(1)

- (ii) $\text{Var}(R) = 1.2$.

(2)

- (b) The number, P , of partially wrapped sweets in a tin is given by

$$P = 3R + 4.$$

Find values for $E(P)$ and $\text{Var}(P)$.

(3)

- (c) The total number of sweets in a tin is 200. Sweets are either correctly wrapped, partially wrapped or unwrapped.

- (i) Express C , the number of correctly wrapped sweets in a tin, in terms of R .

(2)

- (ii) Hence find the mean and variance of C .

(2)

(Total 10 marks)

Q15.

A small corner shop sells packs of sandwiches. The probability distribution for the number of packs sold per day, S , has

$$E(S) = 15 \quad \text{and} \quad \text{Var}(S) = 4.$$

The shop is supplied with 20 packs of sandwiches each day at a cost of £1.00 per pack. Packs are sold at £2.20 each. The shop receives a refund of £0.20 per pack for all those unsold at the end of each day.

- (a) Show that the shop's daily profit, $\mathcal{E}P$, on the sale of packs of sandwiches is given by

$$P = 2S - 16.$$

(3)

- (b) Hence determine the mean and standard deviation of the shop's daily profit on the sale of packs of sandwiches.

(4)

(Total 7 marks)

Q16.

The probability distribution for the number of vehicles, V , involved in each minor accident on a particular stretch of road can be modelled as follows.

v	1	2	3	4	5
$P(V = v)$	0.15	0.45	0.20	0.15	0.05

- (a) Show that $E(V) = 2.5$ and $\text{Var}(V) = 1.15$.

(3)

- (b) The total cost, $\mathcal{E}C$, of removing all the damaged vehicles following a minor accident is given by:

$$C = 30V + 25.$$

Determine the mean and variance of C .

(3)

- (c) The total repair cost, $\mathcal{E}R$, for all the vehicles involved in a minor accident is given by:

$$R = 40V^2 + 15V + 50.$$

Determine the value of $E(R)$.

(3)
(Total 9 marks)

Q17.

Two independent discrete random variables, M and R , are such that $E(M) = 1.7$ and $\text{Var}(M) = 0.81$, with R having the following probability distribution.

r	2	4	6
$P(R = r)$	$\frac{1}{8}$	$\frac{9}{16}$	$\frac{5}{16}$

- (a) Given that M_1 , M_2 , and M_3 are three independent observations on M , write down the value of:

(i) $E(M_1 + M_2 + M_3)$;

(1)

(ii) $\text{Var}(M_1 + M_2 + M_3)$.

(1)

- (b) (i) Calculate values for $E(R)$ and $\text{Var}(R)$.

(4)

- (ii) Hence find values for the mean and variance of $\frac{1}{2}R$.

(3)

- (c) The discrete random variable T is defined by:

$$T = \frac{1}{2}R + M_1 + M_2 + M_3.$$

Evaluate:

(i) $E(T)$;

(1)

(ii) $\text{Var}(T)$.

(2)

(Total 12 marks)

Q18.

The probability distribution for the discrete random variable R is tabulated below.

r	1	2	3	4	5
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$P(R = r)$	0.1	0.2	0.4	0.2	0.1
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(a) Given that $E(R) = 3$ and $\text{Var}(R) = 1.2$, find the mean and variance of $5(2R - 1)$.

(3)

(b) (i) Write down the probability distribution for $\frac{60}{R}$.

(2)

(ii) Show that $E\left(\frac{60}{R}\right) = 24.2$.

(1)

(iii) Given that $E\left(\frac{3600}{R^2}\right) = 759.4$, determine the variance of $\frac{60}{R}$.

(2)

(Total 8 marks)

Q19.

A computer program selects a digit, D , at random from the digits 1, 2, ..., 9.

(a) Explain why $E(D) = 5$

(2)

(b) Given that $E(D^2) = \frac{95}{3}$, find the exact value of $\text{Var}(D)$.

(2)

(c) A number, N , is calculated using

$$N = 3(D + 5).$$

Calculate values for $E(N)$ and $\text{Var}(N)$.

(3)

(Total 7 marks)

Q20.

The discrete random variable R is such that $E(R) = 3$ and $\text{Var}(R) = 1$.

(a) Determine the mean and variance of $5(R - 1)$.

(3)

The probability distribution of R is:

$$P(R = r) = \begin{cases} \frac{r}{10} & r = 1, 2, 3, 4 \\ 0 & \text{otherwise} \end{cases}$$

- (b) Calculate the mean and variance of $12R^{-1}$.

(5)

(Total 8 marks)

Q21.

The discrete random variable T is such that

$$E(T) = 5 \quad \text{and} \quad \text{Var}(T) = 25.$$

- (a) A rectangle has sides of length $2T$ and $(T + 5)$.

Determine the mean and variance of the **perimeter** of the rectangle.

(4)

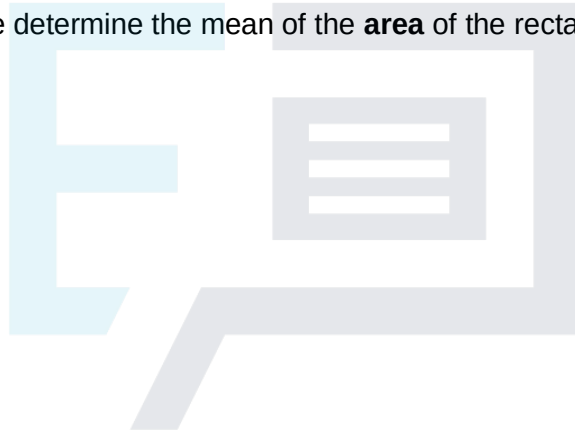
- (b) (i) Show that $E(T^2) = 50$.

(2)

- (ii) Hence determine the mean of the **area** of the rectangle.

(3)

(Total 9 marks)



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